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(54) **DIFFERENTIATED TIME-DIFFERENCE OF
ARRIVAL FOR ULTRA-WIDEBAND (UWB)**

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8/005 (2013.01)

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H04W 8/005; H04W 4/80; H04W 4/02;
H04W 4/025; H04B 17/253; H04B 17/27

See application file for complete search history.

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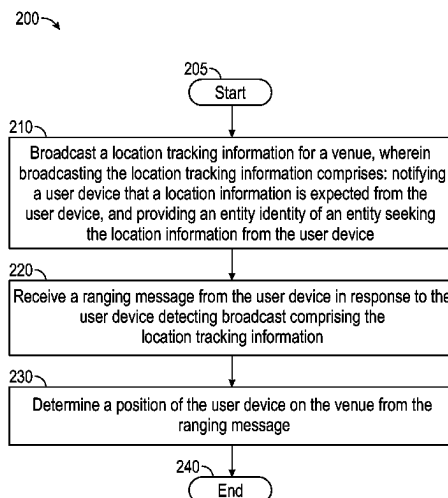
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(57) **ABSTRACT**

Differentiated time-difference of arrival for Ultra-Wideband
(UWB) may be provided. A location tracking information
for a venue may be broadcasted. Broadcasting the location
tracking information may include notifying a user device
that a location information is expected from the user device
and providing an entity identity of an entity seeking the
location information from the user device. A ranging mes-
sage may then be received from the user device in response
to the user device detecting the location tracking informa-
tion. A position of the user device on the venue may be
determined from the ranging message.

15 Claims, 4 Drawing Sheets



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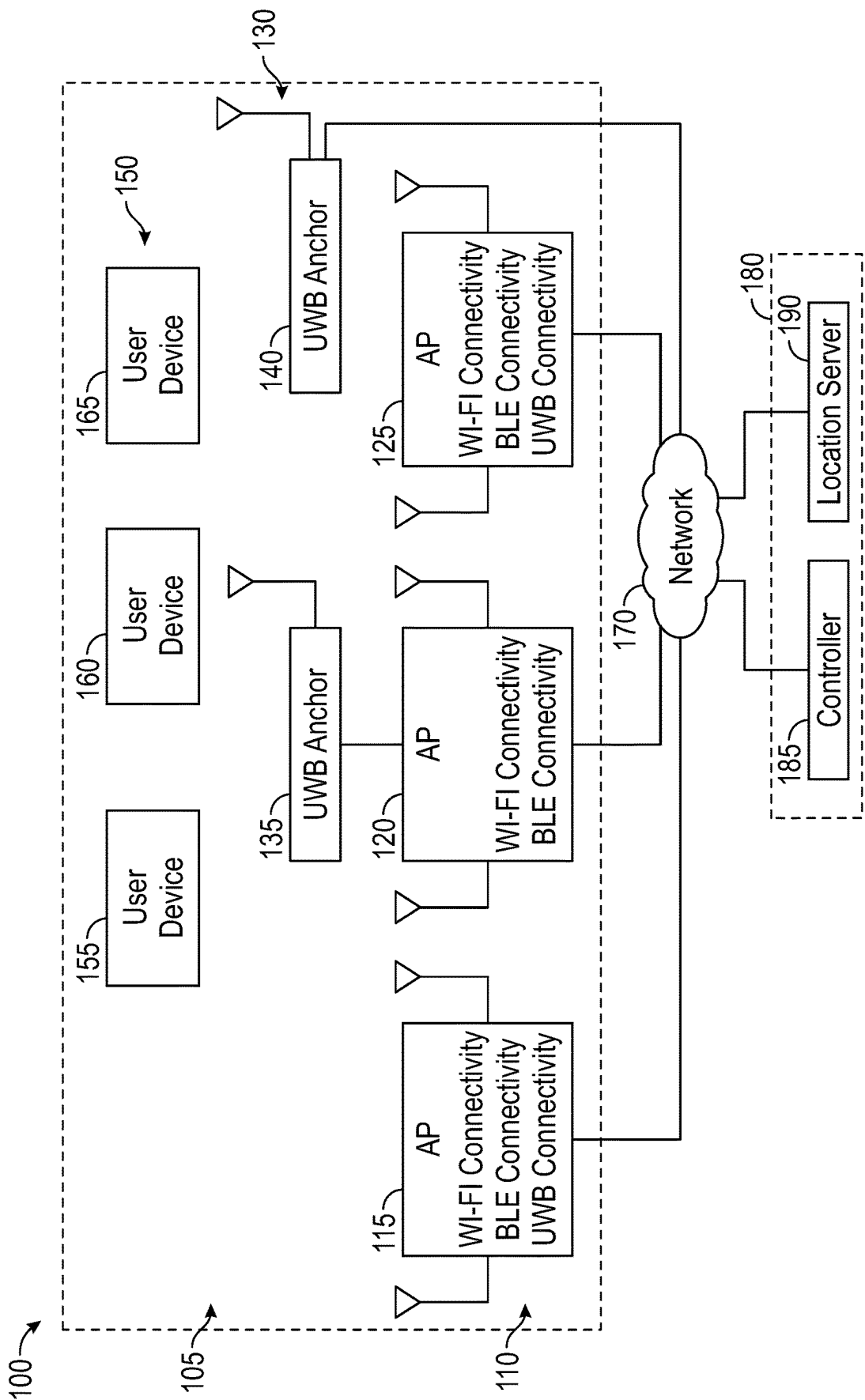


FIG. 1

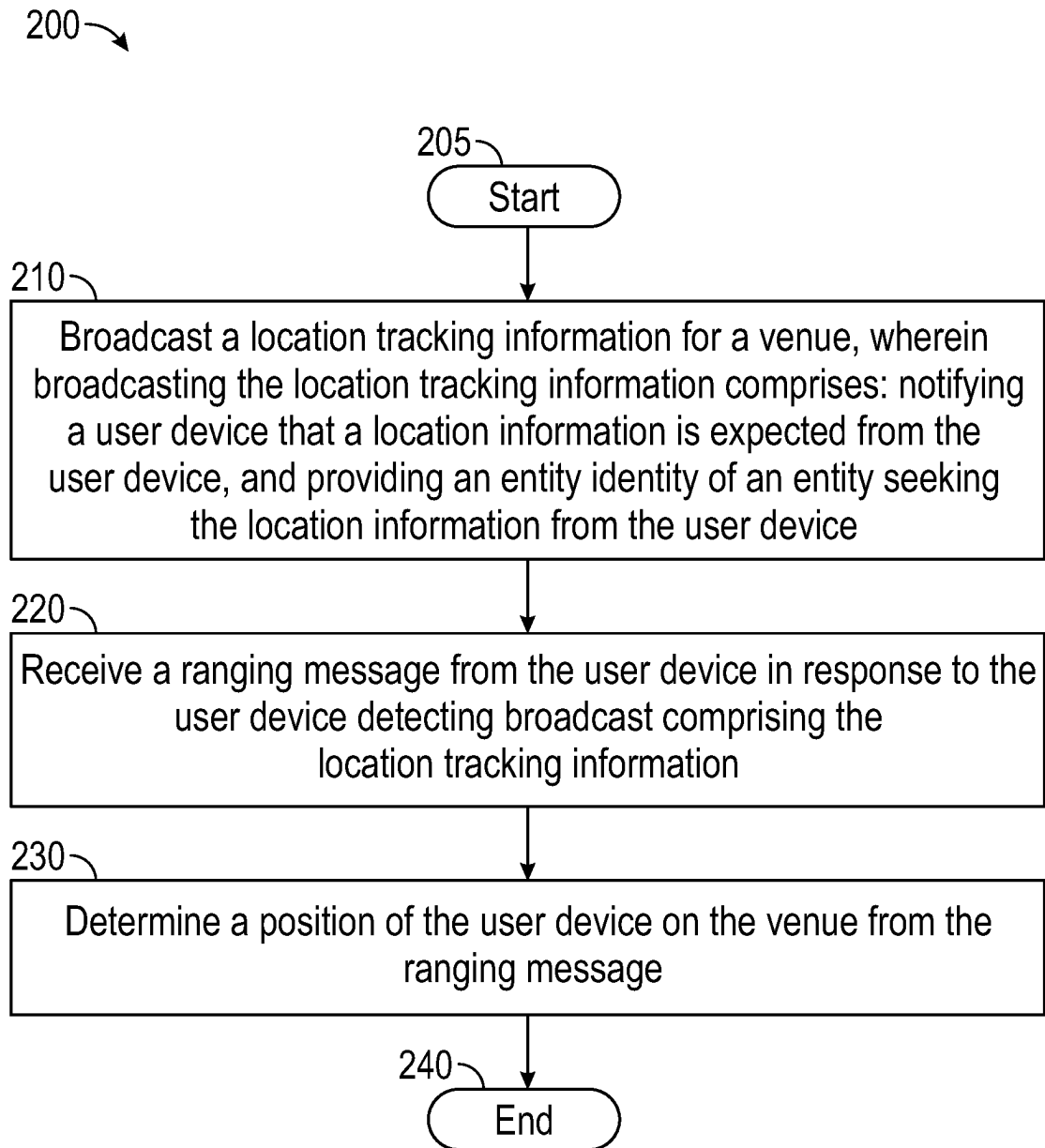


FIG. 2

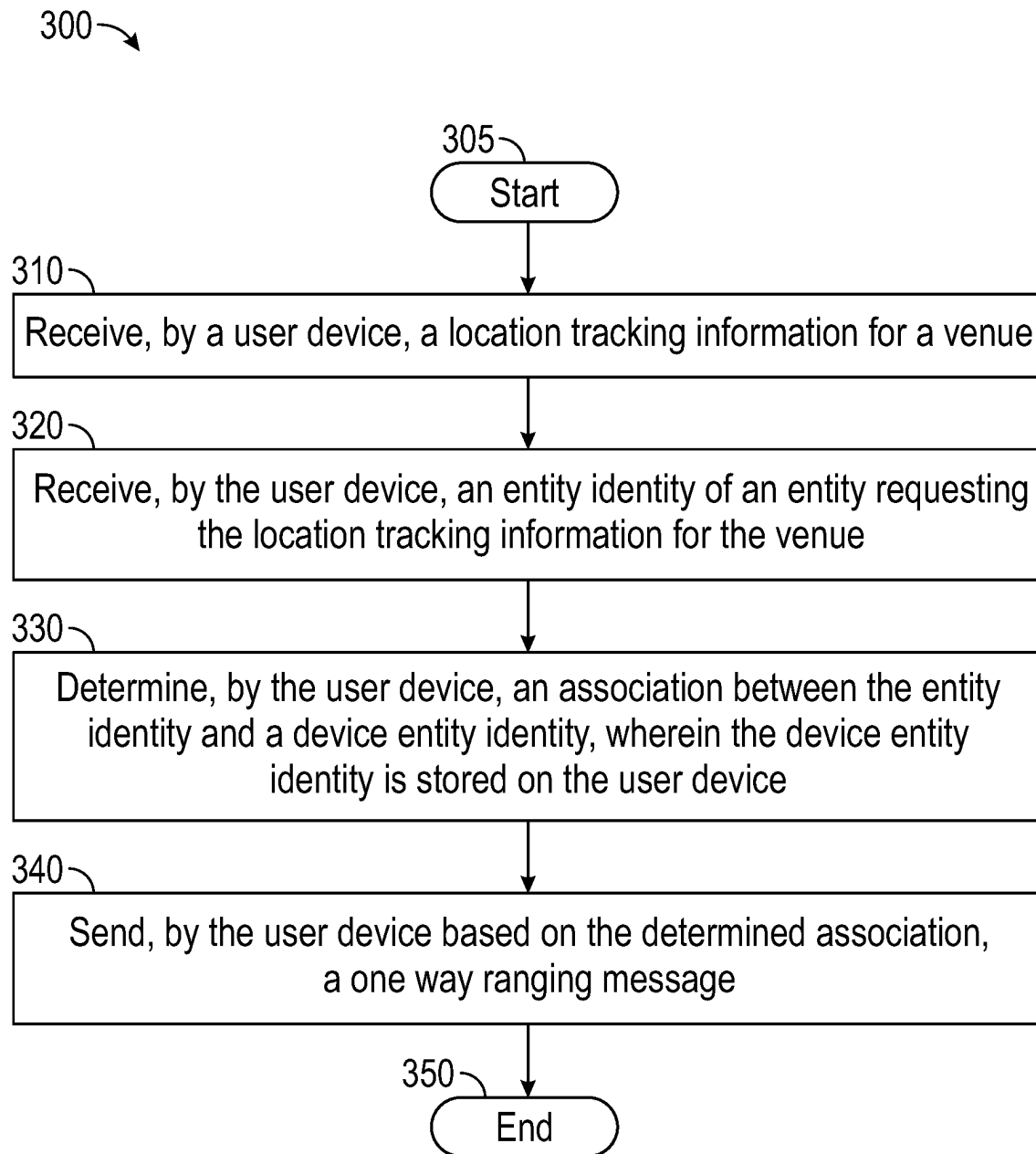
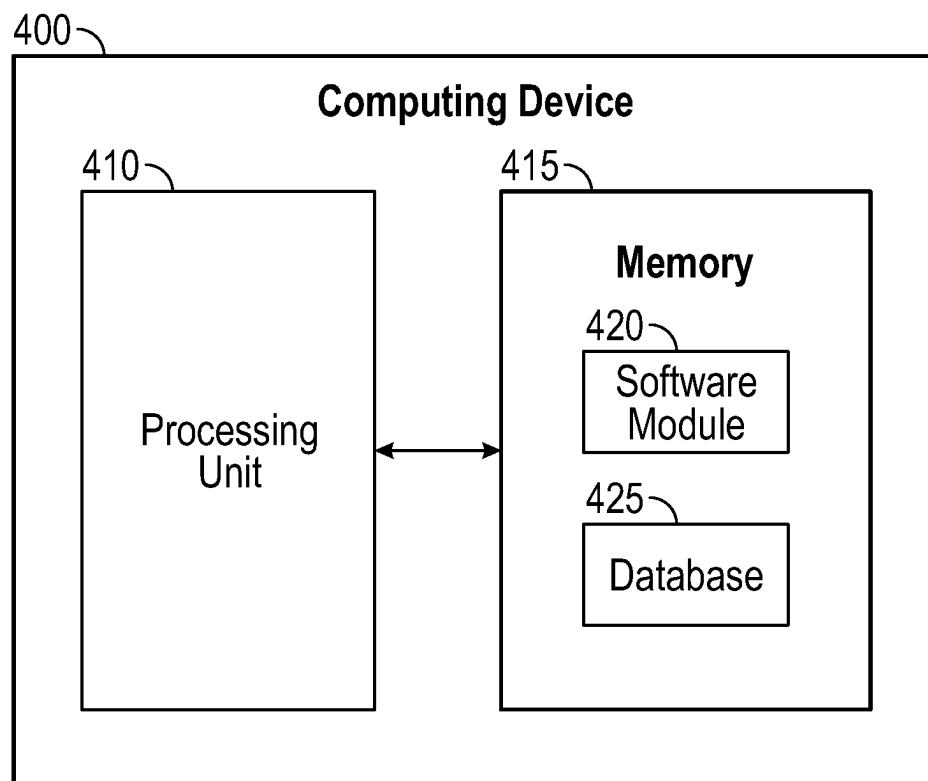


FIG. 3

**FIG. 4**

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**DIFFERENTIATED TIME-DIFFERENCE OF
ARRIVAL FOR ULTRA-WIDEBAND (UWB)****RELATED APPLICATION**

This application claims the benefit of U.S. Provisional Application No. 63/347,327, filed May 31, 2022, and titled “A DIFFERENTIATED TIME-DIFFERENCE OF ARRIVAL FOR ULTRA-WIDEBAND,” the disclosure of which is hereby incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to network equipment and location services.

BACKGROUND

Networking architectures have grown increasingly complex in communications environments, particularly mobile networking environments. In some instances, it is useful to determine a mobile device's location within a mobile networking environment. While the Institute of Electrical and Electronics Engineers (IEEE) 802.11 (e.g., Wi-Fi) or Bluetooth ranging techniques may be utilized in some cases to determine mobile device location, such technologies may provide limited location accuracy. Ultra-Wideband (UWB), as defined in IEEE 802.15.4a and 802.15.4z, may offer improved ranging accuracy over Wi-Fi and Bluetooth, however, utilizing a UWB radio or chip for a mobile device creates a battery cost for the device. Accordingly, there are significant challenges with using UWB ranging for mobile devices.

BRIEF DESCRIPTION OF THE FIGURES

The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various embodiments of the present disclosure. In the drawings:

FIG. 1 is a block diagram of an operating environment for providing a differentiated time-difference of arrival for Ultra-Wideband (UWB);

FIG. 2 is a flow chart of a method for receiving location information from a user device in UWB;

FIG. 3 is a flow chart of a method for providing location tracking information in UWB; and

FIG. 4 is a block diagram of a computing device.

DETAILED DESCRIPTION**Overview**

A differentiated time-difference of arrival for Ultra-Wideband (UWB) may be provided. A location tracking information for a venue may be broadcasted. Broadcasting the location tracking information may comprise notifying a user device that a location information is expected from the user device and providing an entity identity of an entity seeking the location information from the user device. A ranging message may then be received from the user device in response to the user device detecting the location tracking information. A position of the user device on the venue may be determined from the ranging message.

Both the foregoing overview and the following example embodiments are examples and explanatory only and should not be considered to restrict the disclosure's scope, as described, and claimed. Furthermore, features and/or variations may be provided in addition to those described. For

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example, embodiments of the disclosure may be directed to various feature combinations and sub-combinations described in the example embodiments.

EXAMPLE EMBODIMENTS

The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the appended claims.

A fine Ranging specification (e.g., the FiRa specification) may provide a specification for indoor location tracking. The location tracking in the fine ranging specification may be based on Ultra-Wideband (UWB). In one mode, anchors may be organized in clusters, where a primary anchor of each cluster may send Synchronization (Sync) messages at short intervals. All other anchors in the same cluster may respond to the sync messages in turn. All other anchors of the cluster may indicate a Time of Arrival (ToA) and a Time of Departure (ToD) and may synchronize their respective clocks based on the Sync messages. A user device (i.e., a Station (STA)) on the venue may then listen to these messages. The user device may determine its distance relative to each anchor of the cluster using a Downlink (DL) message Time Difference of Arrival (TDoA) computation (also referred to as a DL mode). The user device then may determine its position on the venue using locations of the anchors of the cluster.

In an Uplink (UL) mode, the position of the user device may need to be determined. The user device may send ranging messages (i.e., UL messages). A Real Time Location Service (RTLS) system behind the anchors may use the TDoA of the ranging messages on each anchor to compute a relative distance of the user device to the anchors and thus its position on the venue. However, the fine ranging specification may not provide a process to preserve an identity of the user device in the UL mode. In addition, the fine ranging specification may not provide a process to restrict availability of the DL message for the TDoA computation to authorized devices. Accordingly, embodiments of the disclosure may provide processes to allow a user device to maintain anonymity while sharing location information at a venue. In addition, embodiments of the disclosure may provide processes to force a user device to share location information in a restricted venue or a restricted area of the venue for example.

FIG. 1 shows an operating environment **100** for providing TDoA for UWB. As shown in FIG. 1, operating environment **100** may comprise a venue **105**. Venue **105** may include a plurality of Access Points (APs) **110**, for example, a first AP **115**, a second AP **120**, and a third AP **125**. Venue **105** may further comprise a plurality of UWB anchors **130**, for example, a first UWB anchor **135** and a second UWB anchor **140**. Venue **105** may further comprise a plurality of user devices **150**, for example, a first user device **155**, a second user device **160**, and a third user device **165**. Operating environment **100** may further include a network **170** and a

control device **180**. Control device **180** may include a controller **185** and a location server **190**.

Venue **105** may support any density of plurality of user devices **150**, and may include any indoor or outdoor area, such as a home, school, campus, office building, conference center, stadium, or other venue or location or portion thereof. Venue **105** may include a restricted area or an admission controlled area. Each of plurality of APs **110** may be positioned at known locations in venue **105** and may facilitate a connection between one or more of plurality of user devices **150** and network **170**.

Plurality of APs **110**, for example, may communicate with plurality of user devices **150** through Wi-Fi Wireless Local Area Network (WLAN), Bluetooth Low Energy (BLE), or UWB. For example, plurality of APs **110** may include a Wi-Fi chipset for providing Wi-Fi connectivity, a BLE chipset for providing BLE connectivity, and a UWB chipset for providing UWB connectivity. First AP **115** and third AP **125** may include built-in/integrated Wi-Fi connectivity, BLE connectivity, and UWB connectivity. Second AP **120** may include built-in/integrated Wi-Fi connectivity and BLE connectivity but not UWB connectivity. However, second AP **120** may achieve UWB connectivity via first UWB anchor **135** that may be connected to second AP **120**.

Each of plurality of UWB anchors **130** may also be positioned at known locations in venue **105**, and may receive, send, and process UWB transmissions. Each of plurality of UWB anchors **130** may include other communication capabilities, such as BLE wireless communication capabilities or wired communication capabilities, for example, via a connection to network **170** over IEEE 802.11, Ethernet, or another connection mechanism. First UWB anchor **135** may also be referred to as a peripheral UWB anchor as it may be connected to second AP **120**. Second UWB anchor **140** may also be referred to as a standalone UWB anchor. An anchor, for example, may refer to any device configured to detect UWB transmissions from plurality of user devices **150**. Therefore, each of first AP **115**, third AP **125**, and plurality of UWB anchors **130** may be referred to as an anchor.

Plurality of user devices **150** may comprise, but are not limited to, a smart phone, a personal computer, a tablet device, a mobile device, a telephone, a remote control device, a set-top box, a digital video recorder, an Internet-of-Things (IoT) device, a network computer, a router, an Automated Transfer Vehicle (ATV), a drone, an Unmanned Aerial Vehicle (UAV), or other similar microcomputer-based device.

Each of plurality of user devices **150** may communicate with one or more of plurality of APs **110** and one or more of plurality of UWB anchors **130**. For example, each of plurality of user devices **150** may include Wi-Fi WLAN connectivity for communicating over Wi-Fi WLAN, BLE connectivity for communicating over BLE, and UWB connectivity for communicating over UWB. Plurality of user devices **150** may include a Wi-Fi chipset for providing Wi-Fi connectivity, a BLE chipset for providing BLE connectivity, and a UWB chipset for providing UWB connectivity. Each of plurality of user devices **150** may also include a fine ranging enabled application. The fine ranging enabled application, in an example, may manage UWB connectivity of each of plurality of user devices **150**.

Plurality of APs **110**, plurality of UWB anchors **130**, and plurality of user devices **150** may communicate with control device **180** via network **170**. Network **170** may include any communications medium for transmitting information between two or more computing devices. For example,

network **170** may include a LAN, a Wide Area Network (WAN), a Virtual Private Network (VPN), Intranet, Internet, hardware connections, modem connections, wireless connections, or combinations of one or more these items.

Controller **185** may manage operations of plurality of APs **110**. For example, controller **185** may facilitate communications involving plurality of user devices **150** through plurality of APs **110**. Controller **185** and location server **190** may be separate and physically distinct entities

Location server **190** may manage location-related transmissions involving plurality of APs **110**, plurality of UWB anchors **130**, and plurality of user devices **150**. For example, location server **190** may cooperate with plurality of APs **110**, plurality of UWB anchors **130**, and plurality of user devices **150** to initiate and complete device ranging procedures at venue **105** that provide location measurements to location solutions. Location server **190** may perform location computations, that is, process time, distance, angle, signal strength or other information from one or more of plurality of APs **110**, plurality of UWB anchors **130** and plurality of user devices **150**. Location server **190** may determine or track a position of a particular one of plurality of user devices **150** on venue **105** and may provide to some other entity seeking that location information.

The elements described above of operating environment **100** (e.g., plurality of APs **110**, plurality of UWB anchors **130**, plurality of user devices **150**, control device **180**, controller **185**, and location server **190**) may be practiced in hardware and/or in software (including firmware, resident software, micro-code, etc.) or in any other circuits or systems. The elements of operating environment **100** may be practiced in electrical circuits comprising discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. Furthermore, the elements of operating environment **100** may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including but not limited to, mechanical, optical, fluidic, and quantum technologies. As described in greater detail below with respect to FIG. 4, the elements of operating environment **100** may be practiced in a computing device **400**.

FIG. 2 is a flow chart setting forth the general stages involved in a method **200** consistent with an embodiment of the disclosure for receiving location information from a user device in UWB. Although, method **200** is described with reference to UWB, other wireless ranging processes may be used. Method **200** may be implemented using location server **190** as described in more detail above with respect to FIG. 1. In other embodiments, method **200** may be implemented using one or more of plurality of APs **110** or plurality of UWB anchors **130** as described in more detail above with respect to FIG. 1. Ways to implement the stages of method **200** will be described in greater detail below.

Method **200** may begin at starting block **205** and proceed to stage **210** where location server **190** may broadcast a location tracking information for venue **105**. Broadcasting the location tracking information may comprise notifying first user device **155** that a location information is expected from first user device **155** at venue **105**. In addition, broadcasting the location tracking information may comprise providing an entity identity of an entity seeking the location information from first user device **155**. For example, the fine ranging specification may allow location server **190** to broadcast operational parameters of UWB over Out of Band (OOB) channels (for example, BLE or Wi-Fi) through

plurality of APs **110**. However, the fine ranging specification may limit this advertisement to information that UWB is available in venue **105** for location purposes (along with a UL mode or a DL mode) and Physical (PHY) parameters (e.g., modulation, channel, etc.)

Consistent with embodiments of the disclosure, location server **190** may broadcast a location tracking type information in an information element of a broadcast message. The tracking type information may include a DL mode or a UL mode. For example, the information element of the broadcast message may notify or inform first user device **155** that location information is expected for first user device **155** along with the UL mode. Values in the information element may include, for example, company assets, individual user devices, analytics, controlled areas, lost devices, or others. Location server **190** may also broadcast the entity identity (i.e., a Company ID) value. The entity identity value may be a string representing a venue administrator. The string may be literal (e.g., Company Name) or a code (e.g., OB87DS). The tracking type information may be broadcasted in a BLE signal frequency range or a Wi-Fi signal frequency range.

From stage **210**, where location server **190** broadcasts the location tracking information for venue **105**, method **200** may advance to stage **220** where location server **190** may receive a ranging message from first user device **155** in response to first user device **155** detecting broadcast comprising the location tracking information. The ranging message may be a One Way Ranging (OWR) message and may be received in a UWB channel or in a UWB signal frequency range.

First user device **155** may compare the entity identity received in the location tracking information with a device entity identity. The device entity identity may be stored on first user device **155** on the fine ranging enabled application. If there is match between the received entity identity and the device entity identity, first user device **155** may consider itself to be an asset of the venue administrator and may start sending the OWR messages. The OWR message, when first user device **155** is determined to be an asset of the venue administrator, may include an user device identification information, for example, a Media Access Control (MAC) address of first user device **155**.

If there is no match between the received entity identity and the device entity identity, first user device **155** may determine if the fine ranging enabled application has allowed to share its location. For example, a user associated with first user device **155** may have registered first user device **155** with the entity administrator to share its location in exchange for rewards or coupons in a non-asset mode. The user may have enabled in the fine ranging enabled application to share location in exchange for the rewards in the non-asset mode. In response to determining that the fine ranging enabled application has allowed to share its location, first user device **155** may start sending the OWR messages.

First user device **155** may only allow anonymous location information sharing in the non-asset mode. Anonymous location information sharing may allow location server **190** to determine a general foot traffic pattern at venue **105**, but without user device identification. In the non-asset mode, the fine ranging enabled application on first user device **155** may generate a first random MAC address, a first random timer, and a second random timer. The first random timer may have a first time duration that may, for example, be less than five seconds and the second random time may have a second time duration which may, for example, be between three

minutes and twelve minutes. However, other time durations are possible for both the first random timer and the second random timer.

First user device **155** may send OWR messages using the first random MAC address in the first time duration. At the end of the first time duration or the first timer, first user device **155** may become silent. That is, first user device **155** may not send OWR messages in the second time duration. At the end of the second time duration, first user device **155** may determine if it is still within the UWB range (recognized by the entity identity in the location tracking information broadcasts). If first user device **155** is still within the UWB range at the end of the second time duration, the fine ranging enabled application may generate a new MAC address and new timers. For example, the fine ranging enabled applications may generate a second random MAC address for a third time duration that may begin at the end of the second time duration. The rewards may be received based on a length of the first time duration and the third time duration.

Once location server **190** receives the ranging message from first user device **155** in stage **220**, method **200** may continue to stage **230** where location server **190** may determine a position of first user device **155** on venue **105** from the ranging message. For example, location server **190** may determine the position of first user device **155** on venue **105** based on the TDoA of the ranging messages received by one or more the plurality of anchors of venue **105**. Once location server **190** determines the position of first user device **155** at stage **230**, method **200** may then end at stage **240**.

In some venues **105** (e.g., a restricted area, a hazardous area, etc.), sharing location information may be mandatory or required for first user device **155**. Mandatory location information sharing may be expressed in the information element of broadcasts comprising the location tracking information. Upon detecting a venue mandated location information sharing, the fine ranging enabled application may trigger first user device **155** to send the ranging message with the user device identification information.

FIG. 3 is a flow chart setting forth the general stages involved in a method **300** consistent with an embodiment of the disclosure for providing the location tracking information in UWB. Although, method **300** is described with reference to UWB, other wireless ranging processes may be used. Method **300** may be implemented using any of plurality of user devices **150**, for example, first user device **155**, as described in more detail above with respect to FIG. 1. Ways to implement the stages of method **300** will be described in greater detail below.

Method **300** may begin at starting block **305** and proceed to stage **310** where first user device **155** may receive the location tracking information for venue **105**. First user device **155** may receive the location tracking information from any of plurality of APs **110** in a broadcast message or an information element of the broadcast message. First user device **155** may receive the location tracking information in an OOB signal frequency range, for example, a BLE signal frequency range and a Wi-Fi signal frequency range.

From stage **310**, where first user device **155** receives the location tracking information for venue **105**, method **300** may advance to stage **320** where first user device **155** may receive the entity identity of an entity requesting the location tracking information for venue **105**. First user device **155** may receive the entity identity as an entity identity value in the information element of the broadcast message. The entity identity value may be a string representing a venue administrator. The string may be literal (e.g., Company

Name) or a code (e.g., OB87DS). The tracking type information may be broadcasted in a BLE signal frequency range or a WiFi signal frequency range.

Once first user device **155** receives the entity identity in stage **320**, method **300** may advance to stage **330** where first user device **155** may determine an association between the received entity identity and the device entity identity. For example, first user device **155** may retrieve the device entity identity from the fine ranging enabled application. First user device **155** may then compare the received entity identity with the device entity identity and determine that it is the asset of the venue administrator. In some embodiments, first user device **155** may determine that it is not the asset of the venue administrator, but is registered with the venue administrator. In some other examples, first user device **155** may determine that it is not associated with the venue administrator.

After first user device **155** determines the association between the entity identity and the device entity identity in stage **330**, method **300** may advance to stage **340** where first user device **155** may send the OWR message based on the determined association. For example, in response to determining that it is the asset of the venue administrator, first user device **155** may send the OWR message with the user device identification information. In response to determining that it is not the asset of the venue administrator but is registered with the venue administrator, first user device **155** may send OWR messages without user device identification information. For example, first user device **155** may send OWR messages with a first random MAC address for a first time duration. Once first user device **155** sends the OWR message in stage **340**, method **300** may then end at stage **350**.

In some embodiments, first user device **155** may be lost (i.e., a lost user device). First user device **155** may be reported to be lost to the venue administrator. First user device **155** may also determine that it is lost and activate a lost mode. However, first user device **155** in the lost mode may not know that the ranging message would be detected by another device or the venue administrator. First user device **155** may also weigh a cost of sending the ranging messages to a risk of battery drain in the lost mode. In that case, the venue administrator may activate an OOB lost mode at venue **105**. For example, location server **190** may broadcast the OOB lost mode information at venue **105**. First user device **155** which may be in the lost mode may activate the ranging message transmission in response to detecting broadcast comprising the OOB lost mode. First user device **155** may thus become detectable by the venue administrator.

In venue **105** or in some other venue, the DL mode of UWB may be activated, allowing plurality of user devices **150** to detect the anchor-to-anchor messages and use them to find their position at venue **105**. However, restricted areas may only allow authorized user devices to use the location tracking service in the DL mode. In these restricted areas, the anchor-to-anchor communication may be encrypted, thus obfuscating exchanged timestamps. The obfuscation may leverage a dynamic Scrambled Time Sequence (STS) or another encryption method.

First AP **115** authorized to range in the DL mode may establish a connection with location server **190**, may provide the user device identification information, then may retrieve a series of keys in use for ranging for a next interval at venue **105**. The connection may be performed using Bluetooth connectivity or Wi-Fi connectivity. The series of keys may then be returned over a secure channel and fed into the fine

ranging enabled application of first user device **155**. A non-authorized user device may fail to obtain the series of keys and may be prevented from ranging. The non-authorized user device may be prevented from obtaining the location of plurality of UWB anchors **130** (e.g. 2-layered map, one layer representing the floorplan, and accessible to all, another layer representing plurality of UWB anchors **130** and their location, and accessible only to authorized user devices).

In other embodiments, location server **190** may allow for differential accuracy in the location provider to plurality of user devices **150**. The anchors may exchange a non-encrypted version of their messages, but with a reduced timestamp accuracy (shorter floating point). The encrypted version of the messages may use a more accurate timestamp. Authorized user devices, for example, first user device **155** may therefore compute location with a high level of accuracy (e.g., ten centimeters or less) while other user devices may only obtain general location (e.g., four meters approximation).

FIG. 4 shows computing device **400**. As shown in FIG. 4, computing device **400** may include a processing unit **410** and a memory unit **415**. Memory unit **415** may include a software module **420** and a database **425**. While executing on processing unit **410**, software module **420** may perform, for example, processes for receiving location information from plurality of user devices **150** in UWB as described above with respect to FIG. 2 and processes for providing location tracking information in UWB as described above with respect to FIG. 3. Computing device **400**, for example, may provide an operating environment for plurality of APs **110**, plurality of UWB anchors **130**, plurality of user devices **150**, control device **180**, controller **185**, or location server **190**. Plurality of APs **110**, plurality of UWB anchors **130**, plurality of user devices **150**, control device **180**, controller **185**, and location server **190** may operate in other environments and are not limited to computing device **400**.

Computing device **400** may be implemented using a Wi-Fi access point, a tablet device, a mobile device, a smart phone, a telephone, a remote control device, a set-top box, a digital video recorder, a cable modem, a personal computer, a network computer, a mainframe, a router, a switch, a server cluster, a smart TV-like device, a network storage device, a network relay device, or other similar microcomputer-based device. Computing device **400** may comprise any computer operating environment, such as hand-held devices, multiprocessor systems, microprocessor-based or programmable sender electronic devices, minicomputers, mainframe computers, and the like. Computing device **400** may also be practiced in distributed computing environments where tasks are performed by remote processing devices. The aforementioned systems and devices are examples, and computing device **400** may comprise other systems or devices.

Embodiments of the disclosure, for example, may be implemented as a computer process (method), a computing system, or as an article of manufacture, such as a computer program product or computer readable media. The computer program product may be a computer storage media readable by a computer system and encoding a computer program of instructions for executing a computer process. The computer program product may also be a propagated signal on a carrier readable by a computing system and encoding a computer program of instructions for executing a computer process. Accordingly, the present disclosure may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.). In other words, embodi-

ments of the present disclosure may take the form of a computer program product on a computer-usable or computer-readable storage medium having computer-usable or computer-readable program code embodied in the medium for use by or in connection with an instruction execution system. A computer-usable or computer-readable medium may be any medium that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

The computer-usable or computer-readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, device, or propagation medium. More specific computer-readable medium examples (a non-exhaustive list), the computer-readable medium may include the following: an electrical connection having one or more wires, a portable computer diskette, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, and a portable compact disc read-only memory (CD-ROM). Note that the computer-usable or computer-readable medium could even be paper or another suitable medium upon which the program is printed, as the program can be electronically captured, via, for instance, optical scanning of the paper or other medium, then compiled, interpreted, or otherwise processed in a suitable manner, if necessary, and then stored in a computer memory.

While certain embodiments of the disclosure have been described, other embodiments may exist. Furthermore, although embodiments of the present disclosure have been described as being associated with data stored in memory and other storage mediums, data can also be stored on, or read from other types of computer-readable media, such as secondary storage devices, like hard disks, or a CD-ROM, a carrier wave from the Internet, or other forms of RAM or ROM. Further, the disclosed methods' stages may be modified in any manner, including by reordering stages and/or inserting or deleting stages, without departing from the disclosure.

Furthermore, embodiments of the disclosure may be practiced in an electrical circuit comprising discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. Embodiments of the disclosure may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including but not limited to, mechanical, optical, fluidic, and quantum technologies. In addition, embodiments of the disclosure may be practiced within a general purpose computer or in any other circuits or systems.

Embodiments of the disclosure may be practiced via a system-on-a-chip (SOC) where each or many of the element illustrated in FIG. 1 may be integrated onto a single integrated circuit. Such an SOC device may include one or more processing units, graphics units, communications units, system virtualization units and various application functionality all of which may be integrated (or "burned") onto the chip substrate as a single integrated circuit. When operating via an SOC, the functionality described herein with respect to embodiments of the disclosure, may be performed via application-specific logic integrated with other components of computing device 400 on the single integrated circuit (chip).

Embodiments of the present disclosure, for example, are described above with reference to block diagrams and/or operational illustrations of methods, systems, and computer

program products according to embodiments of the disclosure. The functions/acts noted in the blocks may occur out of the order as shown in any flowchart. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

While the specification includes examples, the disclosure's scope is indicated by the following claims. Furthermore, while the specification has been described in language specific to structural features and/or methodological acts, the claims are not limited to the features or acts described above. Rather, the specific features and acts described above are disclosed as example for embodiments of the disclosure.

The invention claimed is:

1. A method comprising:

broadcasting a location tracking information for a venue, wherein broadcasting the location tracking information comprises:

notifying a user device that a location information is expected from the user device, and

providing an entity identity of an entity seeking the location information from the user device;

receiving a ranging message from the user device in response to the user device detecting broadcast comprising the location tracking information wherein receiving the ranging message from the user device comprises receiving the ranging message comprising a first random Media Access Control (MAC) address for a first time duration;

determining a position of the user device on the venue from the ranging message;

not receiving the ranging message for a second time duration after end of the first time duration; and

receiving the ranging message comprising a second random MAC address for a third time duration after an end of the second time duration.

2. The method of claim 1, wherein broadcasting the location tracking information further comprises broadcasting a location tracking type information comprising an Uplink (UL) mode.

3. The method of claim 1, wherein receiving the ranging message comprises receiving a One Way Ranging (OWR) message from the user device.

4. The method of claim 1, wherein broadcasting the location tracking information comprises broadcasting the location tracking information in an Out of Band (OOB) signal frequency range.

5. The method of claim 4, wherein broadcasting the location tracking information in the OOB signal frequency range comprises broadcasting the location tracking information in at least one of the following: a Bluetooth Low Energy (BLE) signal frequency range and a Wi-Fi signal frequency range.

6. The method of claim 1, wherein receiving the ranging message from the user device comprises receiving the ranging message on an Ultra-Wideband (UWB) signal frequency range.

7. The method of claim 1, wherein receiving the ranging message from the user device comprises receiving the ranging message from the user device wherein the user device is lost.

8. A system comprising:

a memory storage; and

a processing unit coupled to the memory storage, wherein the processing unit is operative to:

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broadcast a location tracking information for a venue, wherein the processing unit being operative to broadcast the location tracking information comprises the processing unit being operative to:

inform a user device that a location information is expected from the user device, and

provide an entity identity of an entity tracking the location information from the user device;

receive a One Way Ranging (OWR) message from the user device in response to the user device detecting broadcast comprising the location tracking information wherein the processing unit being operative to receive the OWR message from the user device comprises the processing unit being operative to:

receive the OWR message comprising a first random Media Access Control (MAC) address for a first time duration;

not receive the OWR message for a second time duration after end of the first time duration; and

receive the OWR message comprising a second random media access control address for a third time duration after end of the second time duration; and

determine a position of the user device on the venue from the OWR message.

9. The system of claim 8, wherein the processing unit being operative to broadcast the location tracking information comprises the processing unit being operative to broadcast a location tracking type information.

10. The system of claim 8, wherein the processing unit being operative to broadcast the location tracking information comprises the processing unit being operative to broadcast the location tracking information in one of the following: a Bluetooth Low Energy (BLE) signal frequency range and a Wi-Fi signal frequency range, and wherein the processing unit being operative to receive the OWR message comprises the processing unit being operative to receive the OWR message from the user device in an Ultra-Wideband (UWB) signal frequency range.

11. A method comprising:

receiving, by a user device, a location tracking information for a venue;

receiving, by the user device, an entity identity of an entity requesting the location tracking information for the venue;

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determining, by the user device, an association between the entity identity and a device entity identity, wherein the device entity identity is stored on the user device; and

sending, by the user device, a One Way Ranging (OWR) message based on the determined association wherein determining the association between the entity identity and the device entity identity comprises determining that the user device is registered with the entity requesting the location tracking information, and wherein sending the OWR message further comprises:

generating a first random Media Access Control (MAC) address in response to determining that the user device is registered with the entity requesting the location tracking information; and

sending the OWR message comprising the first random MAC address for a first time duration;

not sending the OWR message for a second time duration after end of the first time duration;

generating a second random MAC address; and

sending the OWR message comprising the second random MAC address for a third time duration after end of the second time duration.

12. The method of claim 11, wherein:

determining the association between the entity identity and the device entity identity comprises determining that the user device is an asset of the entity requesting the location tracking information; and

sending the OWR message comprising user device identification information in response to determining that the user device is the asset of the entity requesting the location tracking information.

13. The method of claim 11, further comprising:

receiving rewards based on a length of the first time duration.

14. The method of claim 11, wherein the location tracking information is received in one of the following: a Bluetooth Low Energy (BLE) signal frequency range and a Wi-Fi signal frequency range, and wherein the OWR message is sent in an Ultra-Wideband (UWB) signal frequency range.

15. The method of claim 11, wherein the user device is a lost user device.

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